A MINI PROJECT REPORT

ON

LORA BASED MULTIHOP PREDECTIVE ENERGY DEMAND MODELLING FOR WIRELESS COMMUNICATION NETWORKS

Submitted for partial fulfillment of the requirements for the

Award of the degree

of

BACHELOR OF TECHNOLOGY

In

ELECTRONICS AND COMMUNICATION ENGINEERING

By

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

BALAJI INSTITUTE OF TECHNOLOGY AND SCIENCE

Accredited by NBA, NAAC A+ & ISO 9001:2015 Certified Institution

(Affiliated to JNTUH, Hyderabad and Approved by the AICTE, New Delhi)

NARSAMPET, WARANGAL 506 331

2025-26

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CERTIFICATE

This is to certify that this Project Work entitled **"LORA BASED MULTIHOP PREDECTIVE ENERGY DEMAND MODELLING FOR WIRELESS COMMUNICATION NETWORKS"** is a bonafide work carried out by **Neela Varshika (23C3A04C4), Vaka Ravivardhan (23C31A04H1), Sriramula Sharavana (23C31A04G0), Sanga Ravalika (23C31A04F4), Pulishetti Srinivas (23C31A04E3)** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** from **JNTUH, Hyderabad** during the period 2025-26 under our guidance and supervision.

Internal Guide	**Head of the Dept.**
Mr. G. Sandeep	**Dr.P.Kiran Kumar**
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DECLARATION

We, **Neela Varshika (23C3A04C4), Vaka Ravivardhan (23C31A04H1), Sriramula Sharavana (23C31A04G0), Sanga Ravalika (23C31A04F4), Pulishetti Srinivas (23C31A04E3)** hereby declare that the project entitled **"LORA BASED MULTIHOP PREDECTIVE ENERGY DEMAND MODELLING FOR WIRELESS COMMUNICATION NETWORKS"**, submitted in the partial fulfillment of the requirements for the award of B.Tech in Electronics and Communication Engineering to Balaji Institute of Technology & Science, Narsampet, affiliated to JNTUH, Hyderabad is a authentic work and has not been submitted to any other university or institution for award of the degree.

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ABSTRACT

A LoRa-based multi-hop wireless communication network architecture is designed to facilitate the efficient collection and transmission of IoT sensor data, combined with external environmental factors, to enable accurate predictive modeling of energy demand. The multi-hop communication paradigm overcomes the inherent range limitations of single-hop LoRa technology by allowing sensor nodes to forward data through intermediary nodes, thereby expanding network coverage and improving data reliability in geographically dispersed sensor deployments. The network architecture incorporates dynamic routing algorithms to optimize data paths and reduce packet loss, ensuring the integrity of transmitted data critical for energy demand analysis. Comprehensive exploratory data analysis (EDA) is conducted on the collected dataset comprising sensor measurements such as voltage, current, temperature, humidity, and external variables including weather conditions and temporal features. Statistical analysis and visualization techniques such as correlation heatmaps, feature distribution plots, and time series decomposition reveal underlying patterns and feature interactions relevant to energy consumption behaviors. Feature selection based on EDA outcomes enhances model efficiency and reduces computational overhead by eliminating redundant or less significant variables. For predictive classification of energy demand categories, a Support Vector Classifier (SVC) is initially employed due to its capability to handle high-dimensional feature spaces and establish clear decision boundaries. However, to address potential limitations of SVC in capturing complex non-linear relationships and to improve robustness against overfitting, a Random Forest Classifier (RFC) is implemented as the proposed model. RFC's ensemble learning mechanism, which aggregates decisions from multiple decision trees trained on random subsets of data and features, enhances prediction stability and accuracy. Model training and validation employ stratified cross-validation techniques to ensure unbiased performance assessment. Experimental results demonstrate that RFC consistently outperforms SVC across multiple evaluation metrics including accuracy, precision, recall, and F1-score. The RFC model exhibits superior generalization on unseen data, providing more reliable predictions of energy demand fluctuations.

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